c. Gene therapy

d. Diagnosis of infection diseases.

e. To manufacture industrially important products like enzymes using bacteria, fungi and cultured mammalian cells.

12.11 Techniques in Genetic Engineering

There are several techniques used in recombinant DNA technology or gene manipulation. The most frequently used methods are agarose gel electrophoresis, isolation and purification of nucleic acids, nucleic acid blotting techniques, DNA sequencing, chemical synthesis of DNA, gene transfer methods, polymerase chain reaction, construction of gene library, radiolabeling of nucleic acids etc, few of them are discussed here.

12.11.1 Agarose Gel Electrophoresis

Electrophoresis refers to the movement of charged molecules in an electric field. The negatively charged molecules move towards the positive electrode while the positively charged molecules migrate towards the negative electrode. Gel electrophoresis is a routinely used analytical technique for the separation and purification of specific DNA fragments. The gel is composed of either polyacrylamide or agarose. Polyacrylamide gel electrophoresis (PAGE) is used for the separation of smaller DNA fragments while agarose electrophoresis is convenient for the separation of DNA fragments ranging in size from 100 base pairs to 20 kilobase pairs. Gel electrophoresis can also be used for the separation of RNA molecules. A diagramatic view of the agarose gel electrophoresis unit is shown in Figure 12.30a.

Steps

1. Gel is set with wells on one end.
2. The gel is placed in an electrophoresis apparatus and covered with buffer solution.
3. The DNA samples along with tracer dye are placed in the wells of gel.

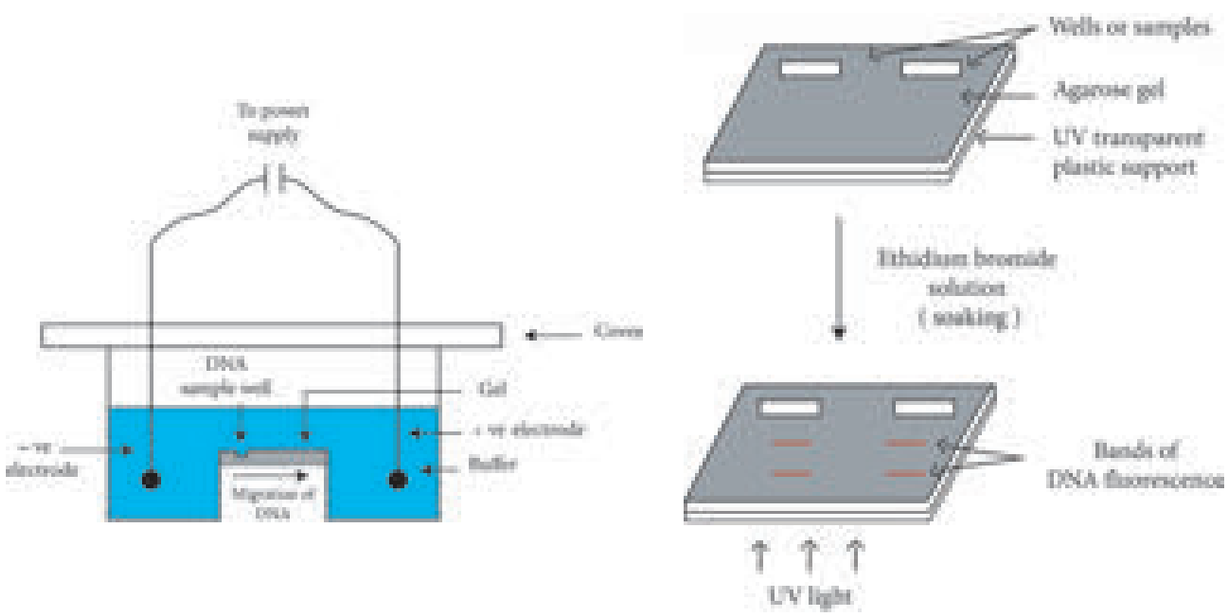


Figure 12.30: **(a)** agarose gel electrophoresis unit **(b)** DNA electrophoresis gel

TN_GOVT_XII_Micro_Biology_CH12.indd 279 04-03-2019 10:51:55